
Introduction to Machine Learning

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About the Course

- **Course Title:** Introduction to Machine Learning
- **Course Code:** CoSc4114
- **Prerequisite:** None
- **Credit Hour:** 3
- **Contact Hours (per week):** Lec – 2 Hrs + 1 Lab

Course Objectives

- At the end of this course, students should be able to:
 - Explain core Machine Learning concepts and types.
 - Prepare and preprocess datasets for modeling.
 - Apply regression and classification algorithms.
 - Apply clustering, and association rule learning techniques.
 - Evaluate model performance using appropriate metrics.
 - Analyze issues such as overfitting and underfitting.
 - Solve real-world problems using Machine Learning.

Course Contents

- **Chapter 1: Introduction**

- What is machine learning?
- History and relationships to other fields
- Types of machine learning techniques
- Applications of machine learning

- **Chapter 2: Supervised learning**

- Linear model
- How to improve supervised models
- Parametric models for classification and regression
- Understand the problems of over-parameterization and the curse of dimensionality
- Use regularization on over-parameterized models

Course Contents ...

- **Chapter 3: Unsupervised learning**
 - Understand the principles of unsupervised learning models
 - Clustering approaches
 - Correctly apply and evaluate clustering models
 - Association rule learning
- **Chapter 4: Neural Network**
 - The Perceptron
 - Training a Perceptron
 - Artificial neural network
 - Multilayer Perceptron
 - Backpropagation algorithm
 - Training procedures
 - Tuning the network size
 - A case study in neural network

Course Contents ...

- **Chapter 5: Model Evaluation**
 - Data processing
 - Data cleaning and transforming
 - Feature selection and visualization
 - Model selection and tuning
 - Methods of dimensional reduction
 - Optimize the performance of the model
 - Over-fitting and Under-fitting
 - Cross-Validation and Re-sampling methods
 - Gradient descent (batch, stochastic)
 - Performance evaluation methods

References

- Bishop, C. M. (2006). *Pattern recognition and machine learning*. Springer-Verlag.
- Shalev-Shwartz, S., & Ben-David, S. (2014). *Understanding machine learning: From theory to algorithms*. Cambridge University Press.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.

Evaluation

- Continuous Assessment 50%
 - Assignments 20%
 - Midterm examination 30%
- Final Examination 50%

Introduction to Machine Learning

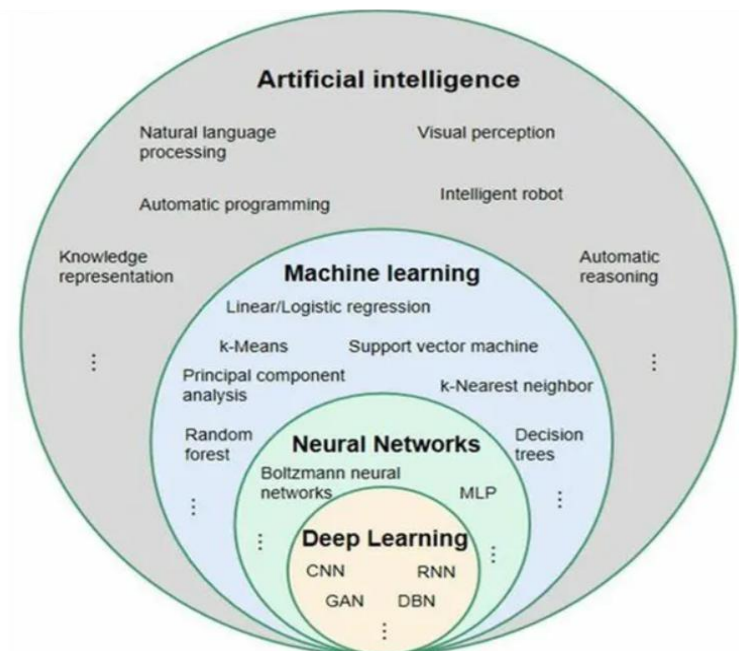
- **Machine Learning (ML)** is a subfield of Artificial Intelligence (AI) that focuses on designing systems that learn from data and improve their performance automatically without explicit programming.
 - ✓ Instead of writing fixed rules, we train models using data.
 - ✓ Systems improve automatically as they are exposed to more data.
- **Key idea:** “Learning from data rather than explicit instructions”

History and Evolution of Machine Learning

- ML has evolved over decades due to advancements in **computing** and **data** availability.
- **Key Phases:**
 1. Early Stage (1950s - 1970s)
 - Basic algorithms like Perceptrons - A simple linear classifier
 - Focus on symbolic AI - Intelligence represented using symbols and rules
 2. Statistical Learning Era (1980s - 1990s)
 - Increased use of probability and statistics
 - Development of algorithms like decision trees and SVMs
 3. Data-Driven Era (2000s - Present)
 - Explosion of big data
 - Growth of deep learning and neural networks
- **Note:** ML evolved from **rule-based systems** -> **data-driven intelligent systems**.

AI vs. ML vs. DL

- **Artificial Intelligence (AI):** Broad field of creating intelligent systems that can perform tasks requiring human intelligence.
- **Machine Learning (ML):** Subset of AI that allows systems to learn from data instead of being explicitly programmed.
- **Deep Learning (DL):** Subset of ML that uses neural networks with many layers.



How Machine Learning Works

- Machine learning typically follows these steps:
 - Data Collection
 - Raw data from sensors, databases, or user input
 - Data Preprocessing
 - Cleaning, normalization, transformation
 - Feature Extraction/Selection
 - Identify important features
 - Model Selection
 - Choose appropriate algorithm (e.g., regression, decision tree, ...)
 - Training the Model
 - Learn patterns from data
 - Evaluation
 - Test model using unseen data
 - Prediction/Deployment
 - Apply model in real-world scenarios

Model = mathematical representation of patterns in data.

Importance of Data in Machine Learning

- **Data** is the backbone of machine learning.
- **Why Data is Important:**
 - Provides examples for learning
 - Determines model accuracy
 - Helps in generalization
 - Enables evaluation and validation
- **Types of Data:**
 - Structured (tables, databases)
 - Unstructured (text, images, audio)

Types of Machine Learning

- Machine learning is broadly categorized into different types:

1) Supervised Learning:

- Learning from labeled data, where input-output pairs are known.
- **Common Tasks:**
 - Classification -> Predict categories (e.g., Email spam detection)
 - Regression -> Predict continuous values (e.g., House price prediction)
- **Common Algorithms:**
 - Linear Regression
 - Logistic Regression
 - Decision Trees, etc.

Types of Machine Learning ...

2) Unsupervised Learning:

- Learning from unlabeled data, discovering hidden patterns.
- **Common Tasks:**
 - Clustering (e.g., Customer segmentation)
 - Association rule learning (e.g., Market basket analysis)
- **Common Algorithms:**
 - K-means clustering, Hierarchical clustering, etc.
 - Apriori Algorithm, FP-Growth, etc

3) Semi-Supervised Learning:

- Combination of labeled and unlabeled data.
- **Use Case:** When labeled data is expensive or limited (Small labeled + Large unlabeled dataset).
- **Benefit:**
 - Improves accuracy when labeling data is expensive.

Types of Machine Learning ...

4) Reinforcement Learning (RL):

- It is a type of ML in which an **agent** learns to make decisions by **interacting** with an environment.
- Unlike supervised learning, the agent is not given correct answers; instead, it learns through **trial** and **error**.
 - The agent performs actions.
 - The environment responds with:
 - Rewards (for good actions)
 - Penalties (for bad actions)
 - The agent improves its behavior over time by aiming to maximize cumulative rewards.
- **Example:**
 - Game-playing AI (**e.g.**, chess, robotics)

Applications of Machine Learning

- Machine Learning is used in many industries to solve problems and improve services. Here are some common real-world applications:
 - **Finance**
 - Fraud detection, Credit scoring, ...
 - **E-commerce**
 - Recommendation systems, Demand forecasting, ...
 - **Healthcare**
 - Disease prediction, Medical image analysis, ...
 - **Natural Language Processing (NLP)**
 - Language translation, Chatbots, Sentiment analysis, ...
 - **Computer Vision**
 - Face recognition, Object detection, ...
 - **Transportation**
 - Self-driving cars, ...

Advantages of Machine Learning

- Handles large-scale data efficiently
- Improves automatically with experience
- Reduces need for manual programming
- Enables intelligent automation

Challenges of Machine Learning

- **Data Issues**

- Poor quality data leads to poor models
- Bias in data causes unfair predictions

- **Model Complexity**

- Difficult to interpret (black-box models)

- **Privacy & Security**

- Sensitive data risks

- **Overfitting & Underfitting**

- Overfitting -> model memorizes data
- Underfitting -> model fails to learn

❖ Ethical concerns are critical in real-world ML systems.

Future of Machine Learning

- Machine learning continues to evolve with trends such as:
 - Deep learning advancements
 - Explainable AI (XAI) -> Make ML decisions understandable to humans
 - Integration with IoT and Big Data
 - Automated Machine Learning (AutoML)
 - Automation of:
 - Model selection
 - Feature engineering
 - Hyperparameter tuning
- ❖ Future systems will be more intelligent, autonomous, and human-aware.

Thank You!